

mk.72-1st hlf 13-K

Con. 6572-13.

GS-6525

(3 Hours)

[Total Marks : 100

N.B. (1) Question No. 1 is **compulsory**.(2) Attempt any **four** questions from remaining **six** questions.(3) **Figures** to the **right** indicate **full** marks.(4) Assume suitable **data** if **necessary**.

1. (a) Prove by Mathematical Induction —

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$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

(b) Explain the terms :—

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(i) Poset

(ii) Normal Subgroup

(iii) Lattice.

(c) In a survey of 60 people, it was found that 25 reads Newsweek Magazine, 26 reads Times and 26 reads Fortune. Also 9 reads both Newsweek and Fortune, 11 reads both Newsweek and Times, 8 reads Time and Fortune and 8 reads no magazine at all.

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(i) Find the number of people who read all three magazines.

(ii) Determine number of people who read exactly one magazine.

2. (a) Define injective, surjective and bijective functions.

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if $f : R \rightarrow R$ and $g : R \rightarrow R$ are defined by the formulas —

$$f(x) = x + 2 \text{ and } g(x) = x^2$$

Find (i) f.g.f. (ii) g.f.g.

(b) Define equivalence relation on a set. Let R be a relation on the set of integers defined by aRb iff $a-b$ is a multiple of 5. Prove that R is a equivalence relation.

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(c) State the converse, inverse and contrapositive of the following :—

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(i) If it is cold, then he wears a hat.

(ii) If an integer is a multiple of 2, then it is even.

3. (a) Explain Hasse diagram. Draw the Hasse diagram of the relation given by :—

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$$(i) R_1 = \{ (1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (2, 2), (2, 3), (2, 4), (2, 5), (3, 3), (3, 4), (3, 5), (4, 4), (4, 5), (5, 5) \}$$

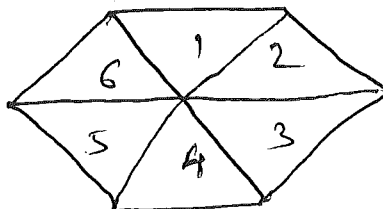
$$(ii) R_2 = \{ (1, 1), (1, 3), (1, 4), (1, 5), (2, 2), (2, 3), (2, 4), (2, 5), (3, 3), (3, 4), (3, 5), (4, 4), (5, 5) \}$$

(b) Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 2), (2, 3), (3, 4), (2, 1)\}$. Find the Transitive closure of R using Warshall's Algorithm.

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(c) Consider the region shown below. It is bounded by a regular hexagon whose sides are the length 1 units. Show that if any seven points are chosen in this region then two of them must be no further apart than 1 unit.

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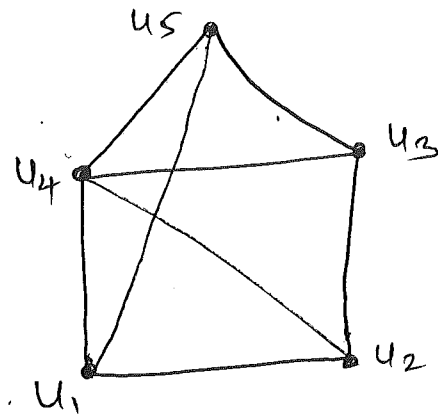
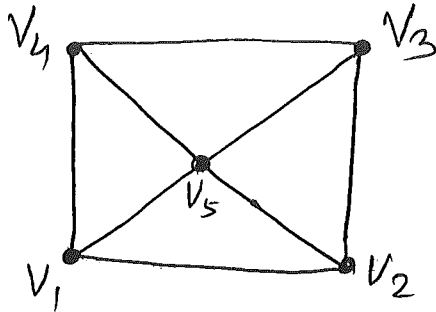
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Con. 6572-GS-6525-13.

2

4. (a) Show that the following graphs are isomorphic.

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(b) Let $R = \{(1, 2), (4, 3), (2, 2), (2, 1), (3, 1)\}$ be a relation on $s = \{1, 2, 3, 4\}$. Find the symmetric closure of R . 6

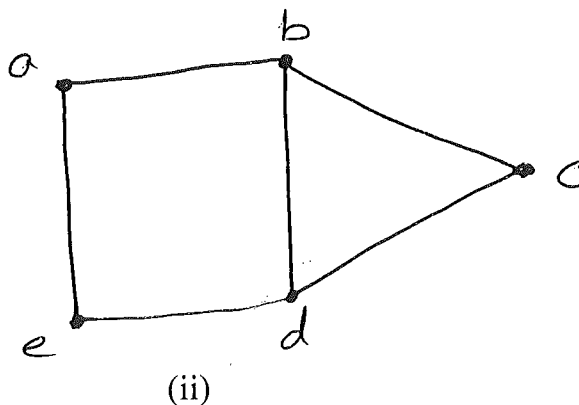
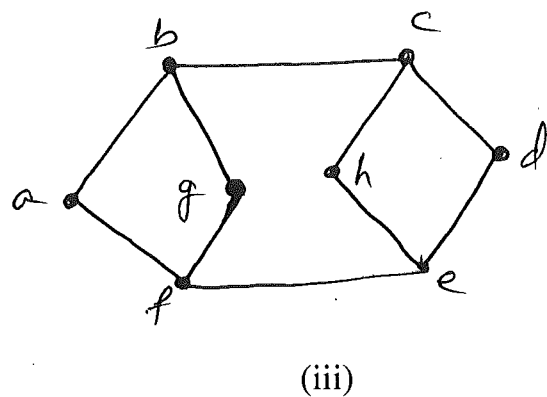
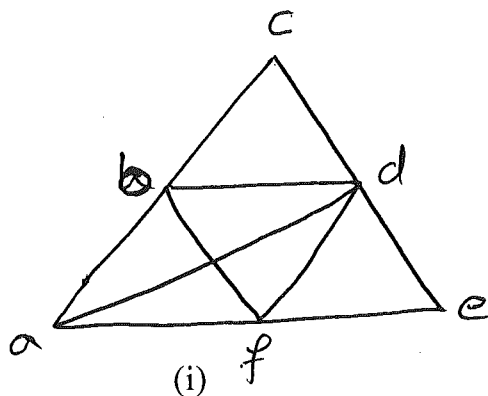
(c) Define : 6

- (i) Integral domain
- (ii) Field
- (iii) Normal Subgroup.

5. (a) What is a minimum spanning tree? Explain any one technique with example. 8

(b) Define Cyclic Group. Prove that the set $A = \{0, 1, 2, 3, 4, 5\}$ is a finite abelian under addition modulo 6. 6

(c) Determine whether the given graph has a Hamilton circuit or Eulerian circuit. If it does, find such a circuit. 6



Con. 6572-GS-6525-13.

3

6. (a) Consider the $(3, 5)$ group encoding function $E : B^3 \rightarrow B^5$ is defined by —

$$E(000) = 00000 \quad E(100) = 10011$$

$$E(001) = 00110 \quad E(101) = 10101$$

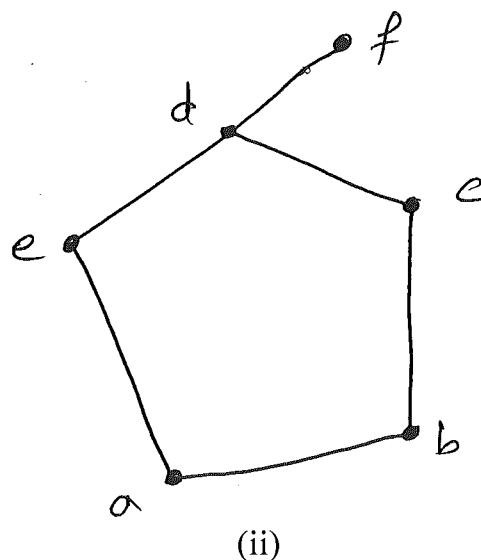
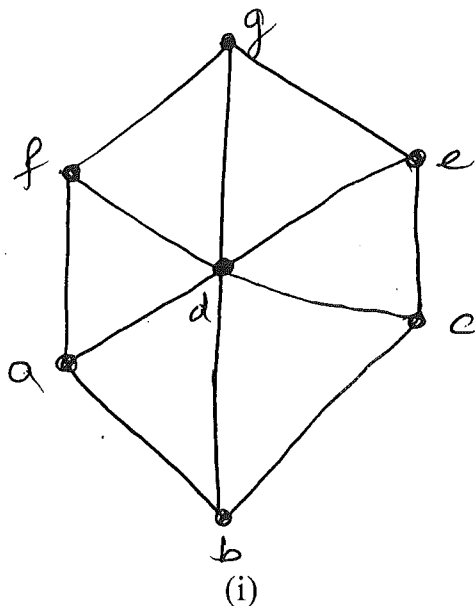
$$E(010) = 01001 \quad E(110) = 11010$$

$$E(011) = 01111 \quad E(111) = 11100$$

Decode the following words relative to a maximum decoding function :—

(i) 11001 (ii) 01010 (iii) 00111 (iv) 11100

(b) Which of the following diagram in the **figure** represents a lattice ? Justify.



(c) Define with example :—

(i) Planer graph

(ii) Semigroup

(iii) Quantifiers.

7. (a) Find the solution to the recurrence relation :—

$$a_n = a_{n-1} + 2 \quad n \geq 2$$

subject to initial condition $a_1 = 3$.

(b) Find the complement of each element in D_{30} .

(c) Find the generating function for each of the following sequence :—

(i) $\{0, 1, 2, 3, 4, \dots\}$

(ii) $\{1, 2, 3, 4, \dots\}$

(iii) $\{2, 2, 2, 2, \dots\}$

(iv) $\{0, 0, 1, 1, 1, \dots\}$